Student Name: Nguyen Bao Chinh

Student ID: 24520225

Link GitHub: https://github.com/Pengikas/24520225_NguyenBaoChinh_Lab_EX-HW.git

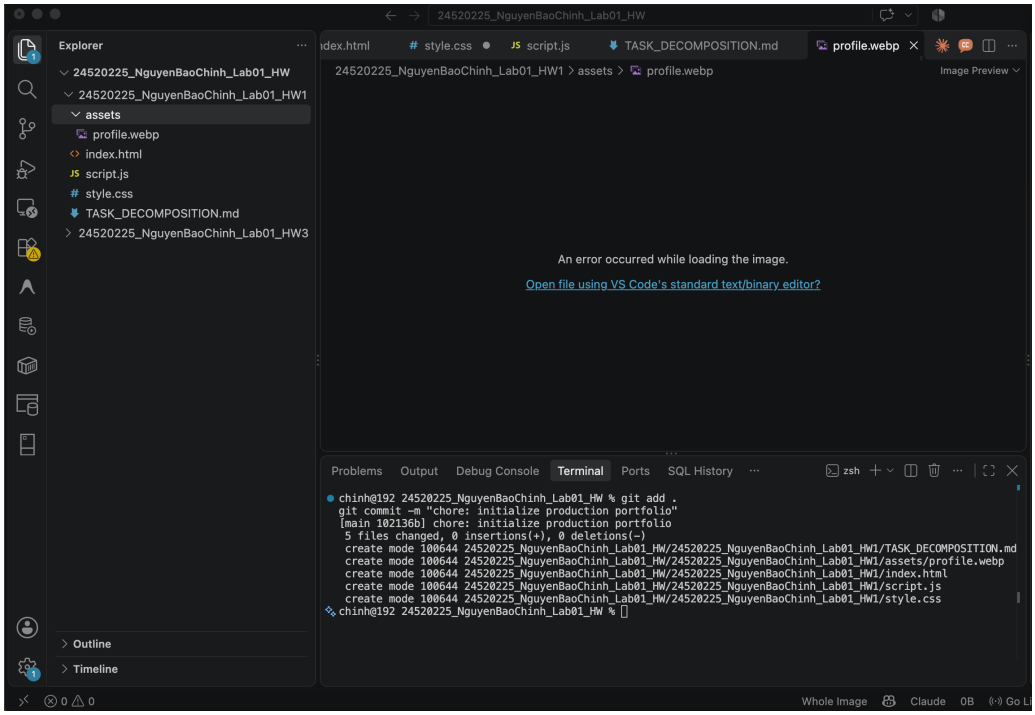
Link AI: https://chatgpt.com/c/6ac5073a-2618-83ec-811d-8f11ce530052, https://chatgpt.com/share/6ac523fd-6e48-83ec-9462-20553e42d566

EVIDENCE FOR LAB01 – HOMEWORK

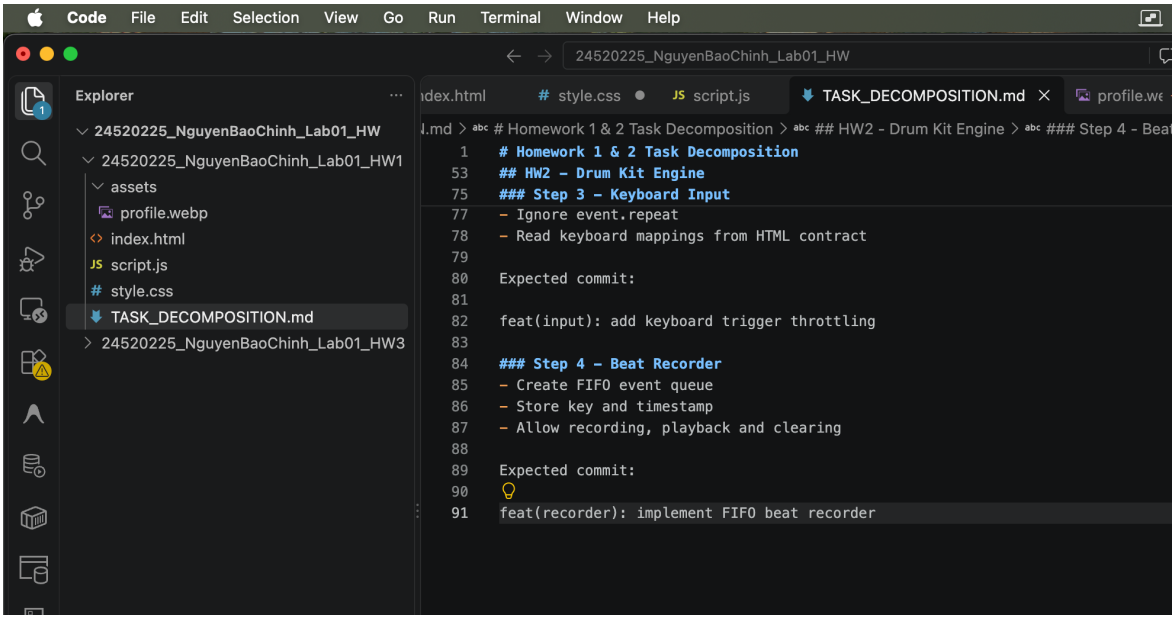
I. Exercise 1: MANDATORY ATOMIC MILESTONES (TASK_DECOMPOSITION.md):

1. M1: WCAG 2.2 AA audit -> Commit 'fix(a11y): contrast & landmarks'

a. Initialize production portfolio



b. Add homework task decomposition



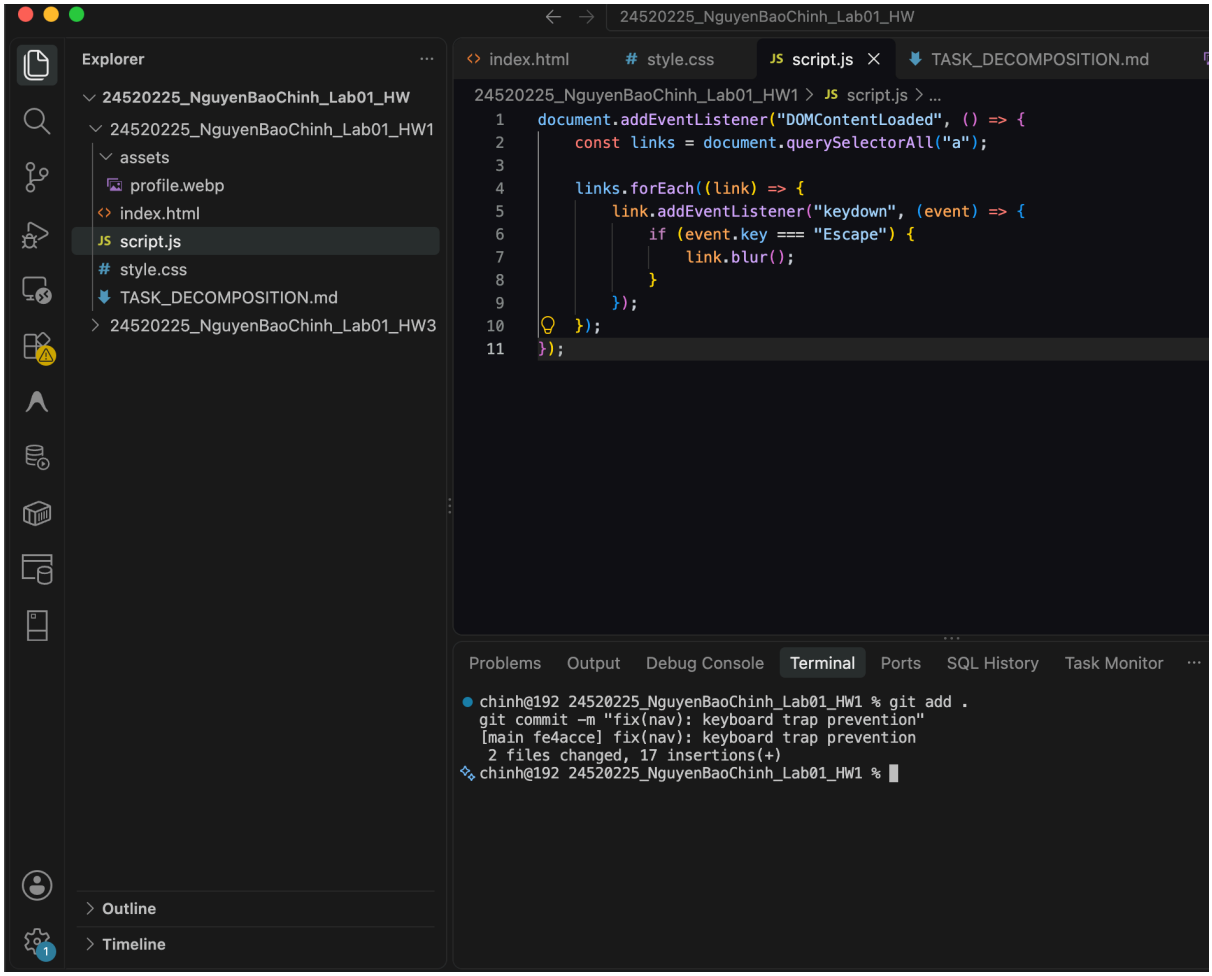

```
chinh@192 24520225_NguyenBaoChinh_Lab01_HW1 % git add TASK_DECOMPOSITION.md
git commit -m "docs: add homework task decomposition"
[main 5dea298] docs: add homework task decomposition
 1 file changed, 91 insertions(+)
```

c. **Commit M1**

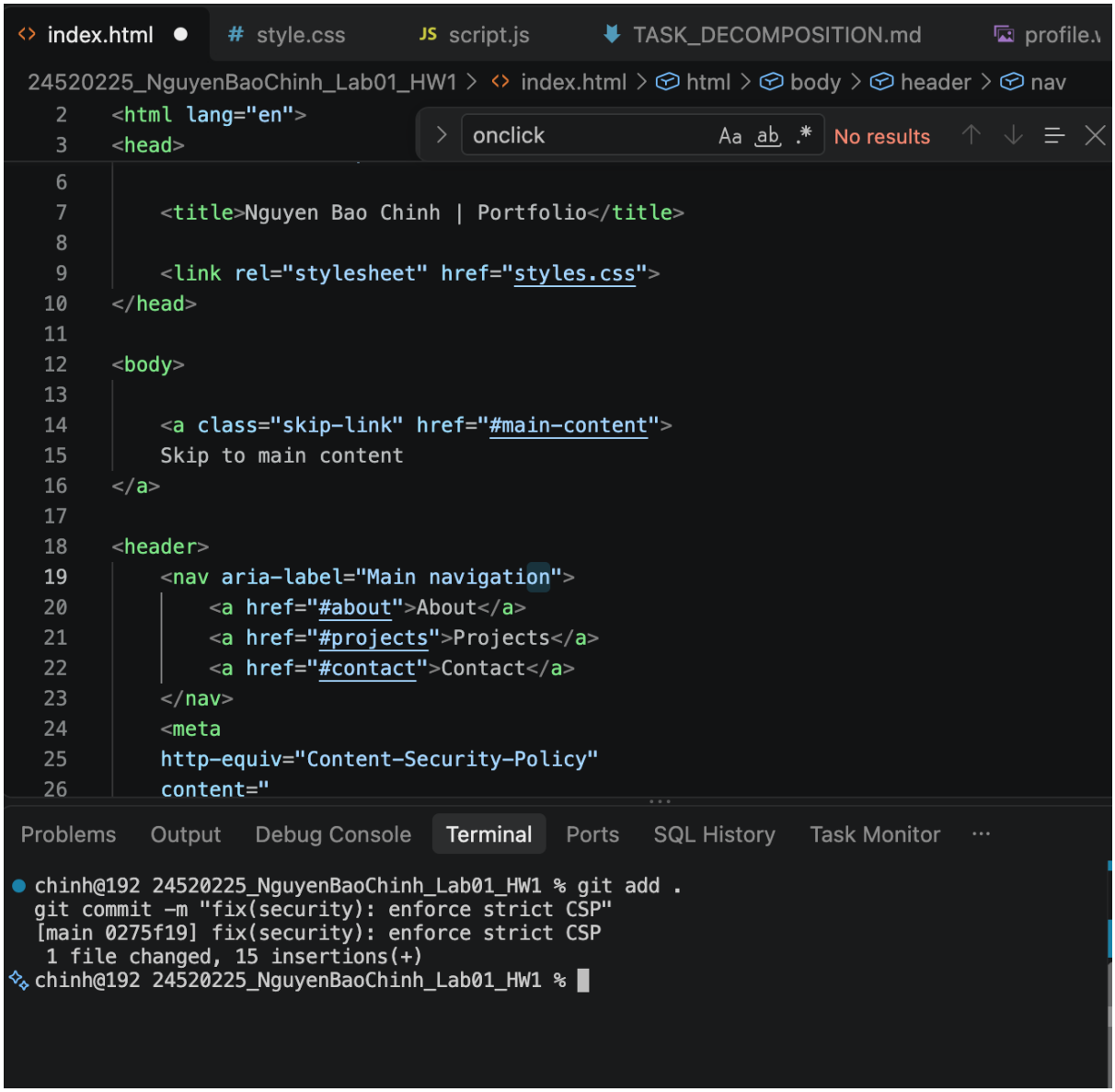
```
chinh@192 24520225_NguyenBaoChinh_Lab01_HW1 % git add index.html styles.css
git commit -m "fix(a11y): contrast & landmarks"
fatal: pathspec 'styles.css' did not match any files
[main a000cdb] fix(a11y): contrast & landmarks
 4 files changed, 238 insertions(+)
 rename README.md => 24520225_NguyenBaoChinh_Lab01_EX/README.md (100%)
 create mode 100644 24520225_NguyenBaoChinh_Lab01_EX/styles.css
```

24520225_NguyenBaoChinh_Lab01_HW/24...	fix(a11y): contrast & landmarks	1 minute ago

2. **M2: Focus trap audit -> Commit 'fix(nav): keyboard trap prevention'**



3. M3: Strict CSP & zero inline handlers (onclick=... is strictly banned)



4. **M4: Lighthouse 100 audit score -> Commit 'perf: optimize assets'**

```
chinh@192 24520225_NguyenBaoChinh_Lab01_HW1 % git add .
git commit -m "perf: optimize assets"
[main 38f77c3] perf: optimize assets
```

5. **Result of M1**

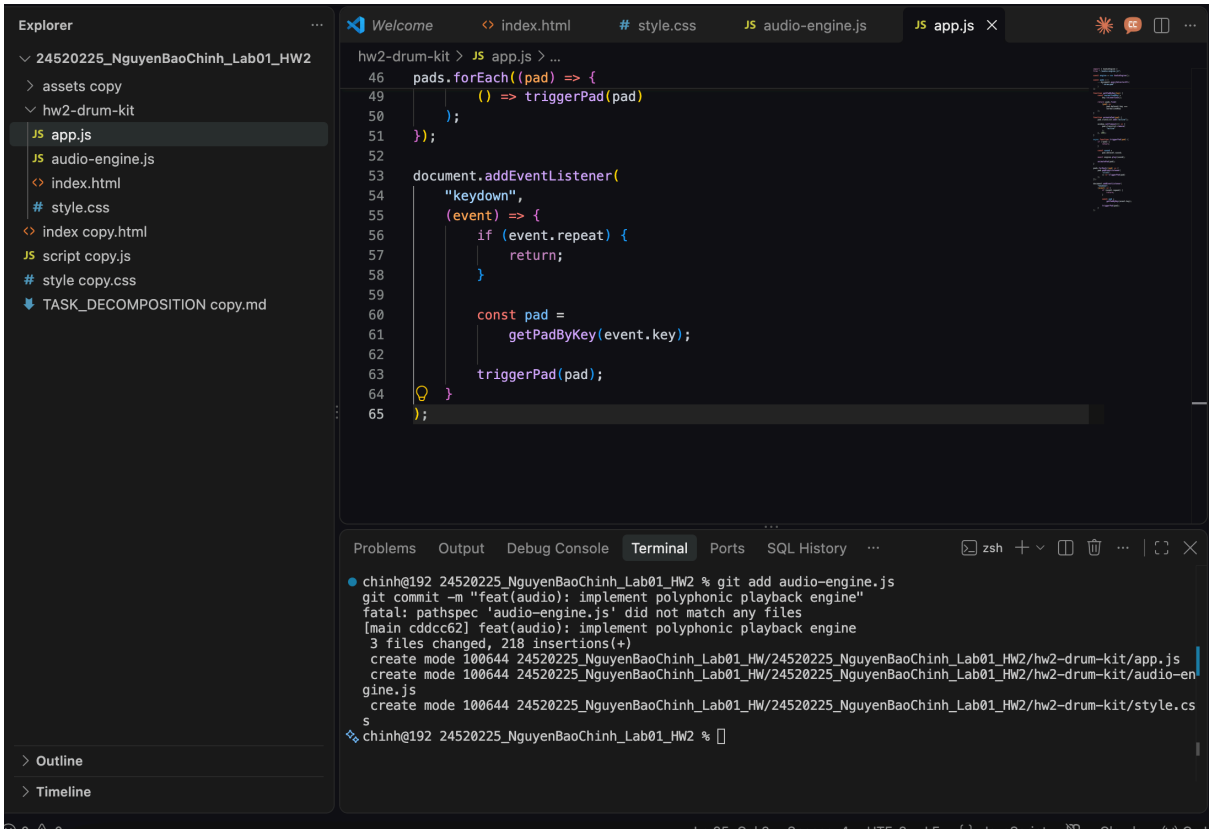
```
chinh@192 24520225_NguyenBaoChinh_Lab01_HW1 % git log --oneline
38f77c3 (HEAD -> main) perf: optimize assets
0275f19 fix(security): enforce strict CSP
fe4acce fix(nav): keyboard trap prevention
a000cdb (origin/main, origin/HEAD) fix(a11y): contrast & landmarks
5dea298 docs: add homework task decomposition
102136b chore: initialize production portfolio
3b598b1 feat(js): skeleton
```

II. Exercise 2: Drum Kit Engine (Contract-First)

1. **Step 1: Commit HTML data-sound contract before touching JavaScript.**

```
chinh@192 24520225_NguyenBaoChinh_Lab01_HW2 % git add index.html
git commit -m "feat(drum): define HTML sound contract"
fatal: pathspec 'index.html' did not match any files
[main 9bc30f5] feat(drum): define HTML sound contract
```

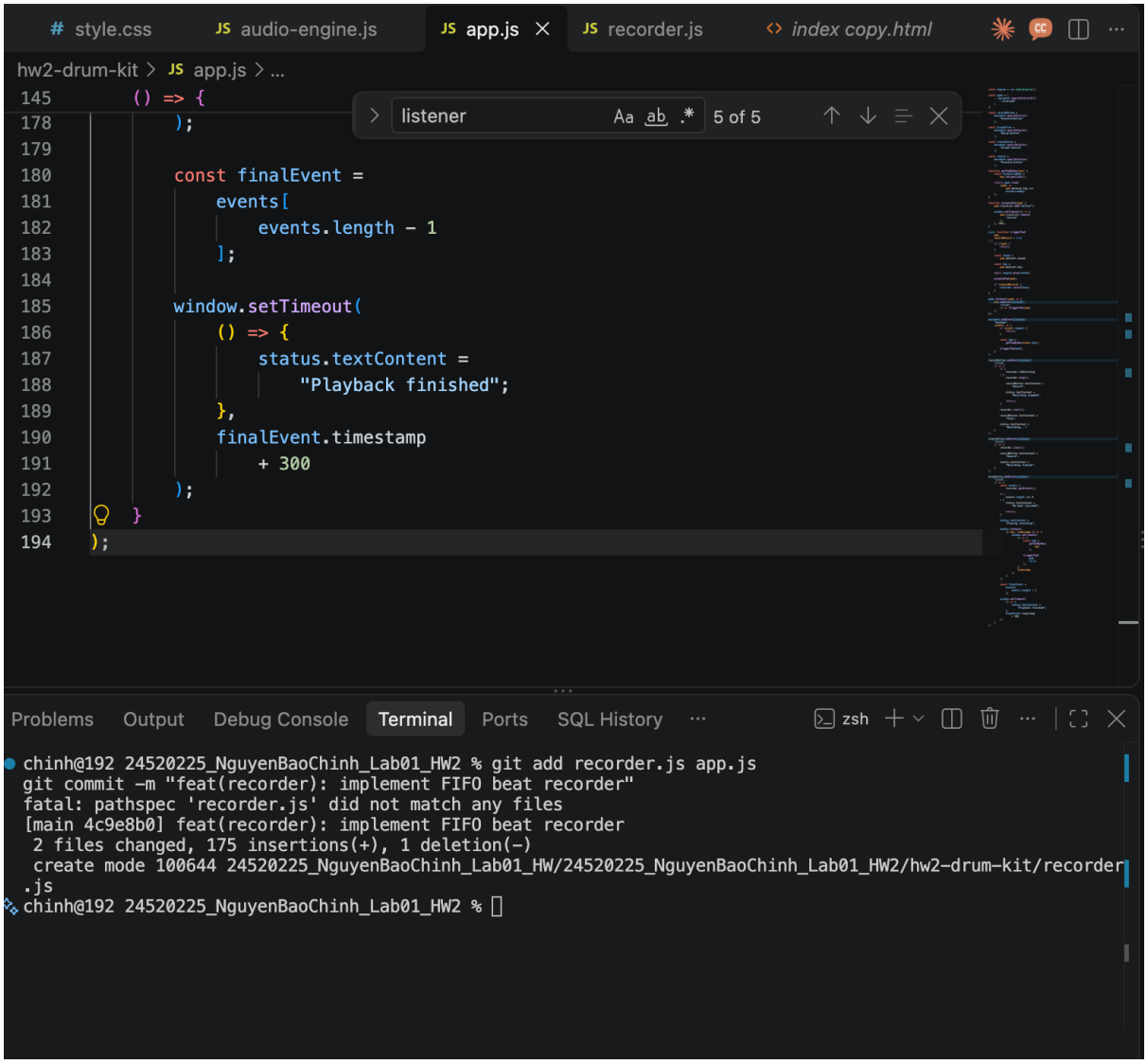
2. **Step 2: Implement polyphonic Audio playback engine independently.**



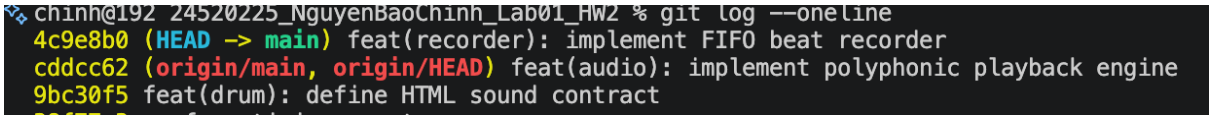
3. **Step 3: Implement keydown listener with event.repeat throttling.**

```
chinh@192 24520225_NguyenBaoChinh_Lab01_HW2 % git add app.js styles.css
git commit -m "feat(input): add keyboard trigger throttling"
fatal: pathspec 'app.js' did not match any files
[main c12c628] feat(input): add keyboard trigger throttling
 1 file changed, 1 insertion(+), 1 deletion(-)
```

4. **Step 4: Implement FIFO Beat Recorder (timestamped event queue).**



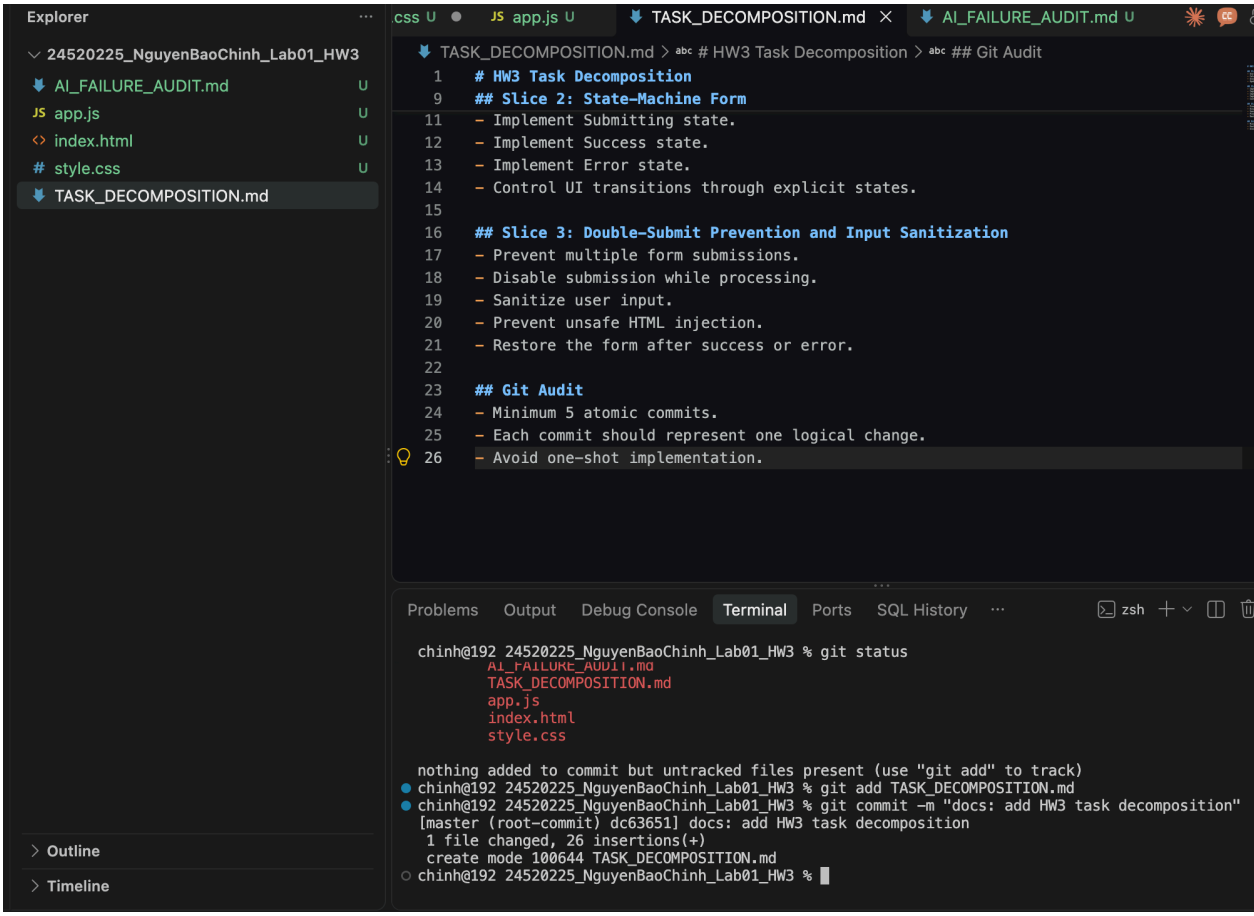
5. **Result**



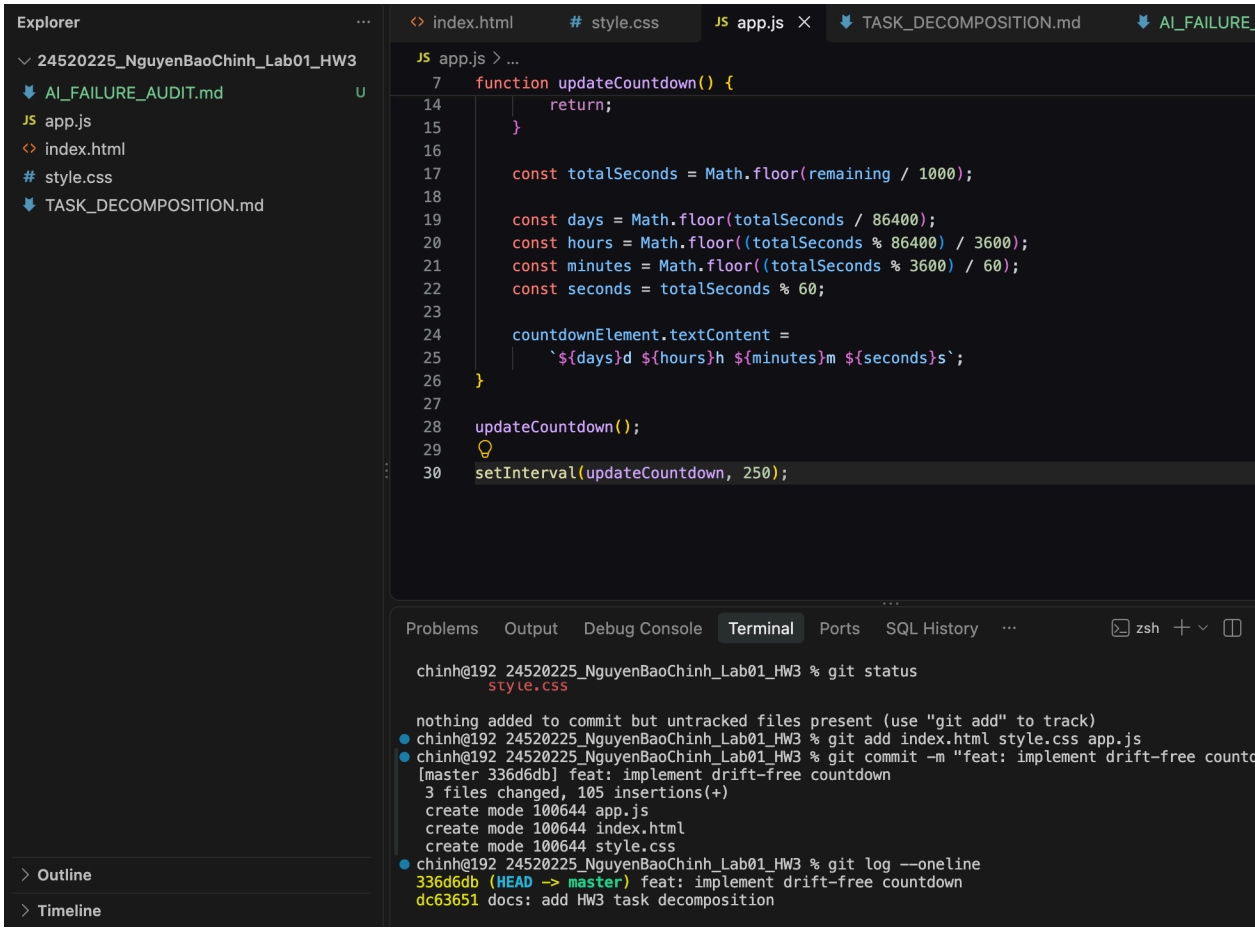
III. Exercise 3: Resilient Landing Page (3 Slices

1. SLICE 1: Drift-Free Countdown Engine (UTC ISO 8601 timestamps).

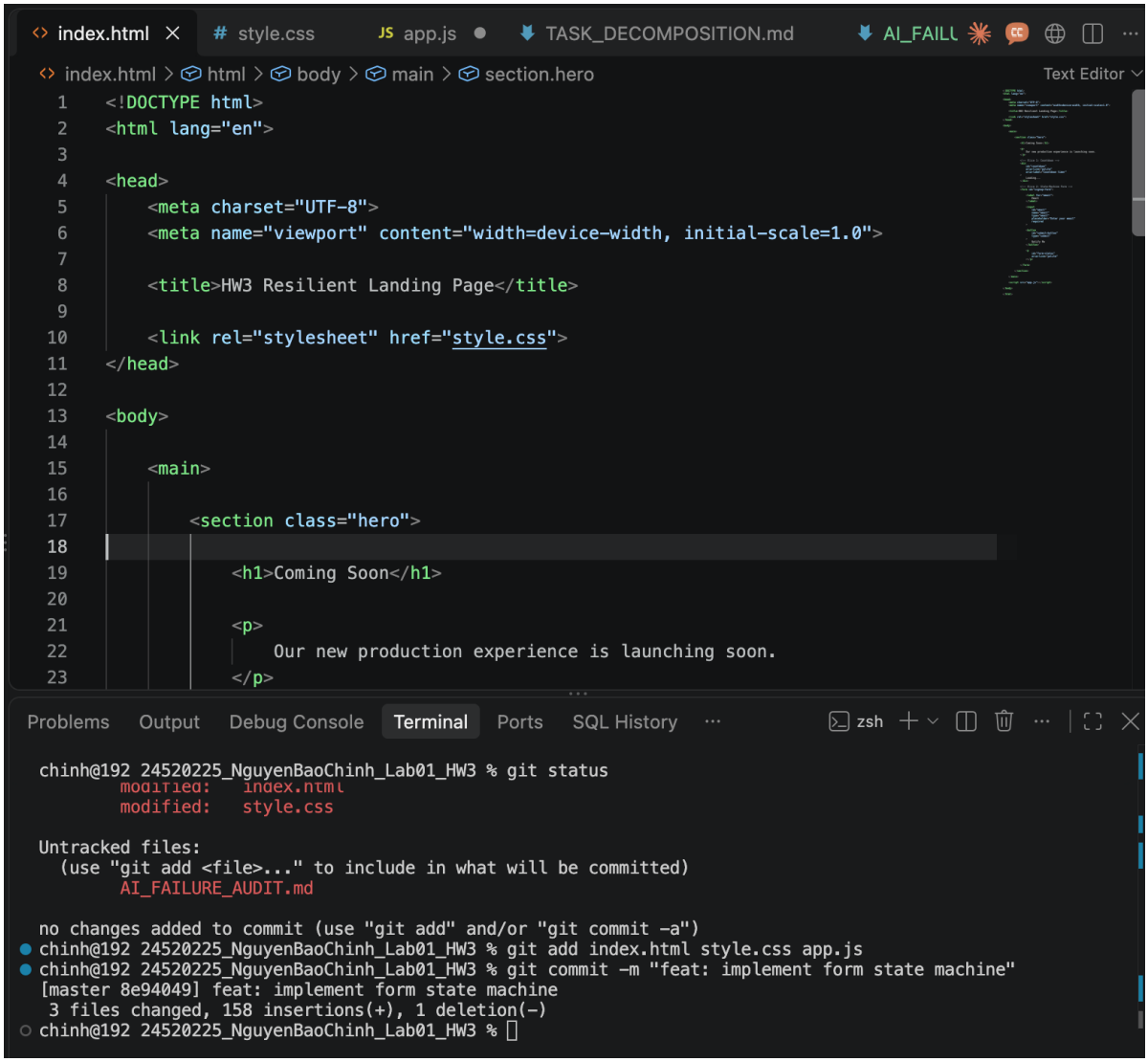
a. Add HW3 task decomposition



b. Implement drift-free countdown



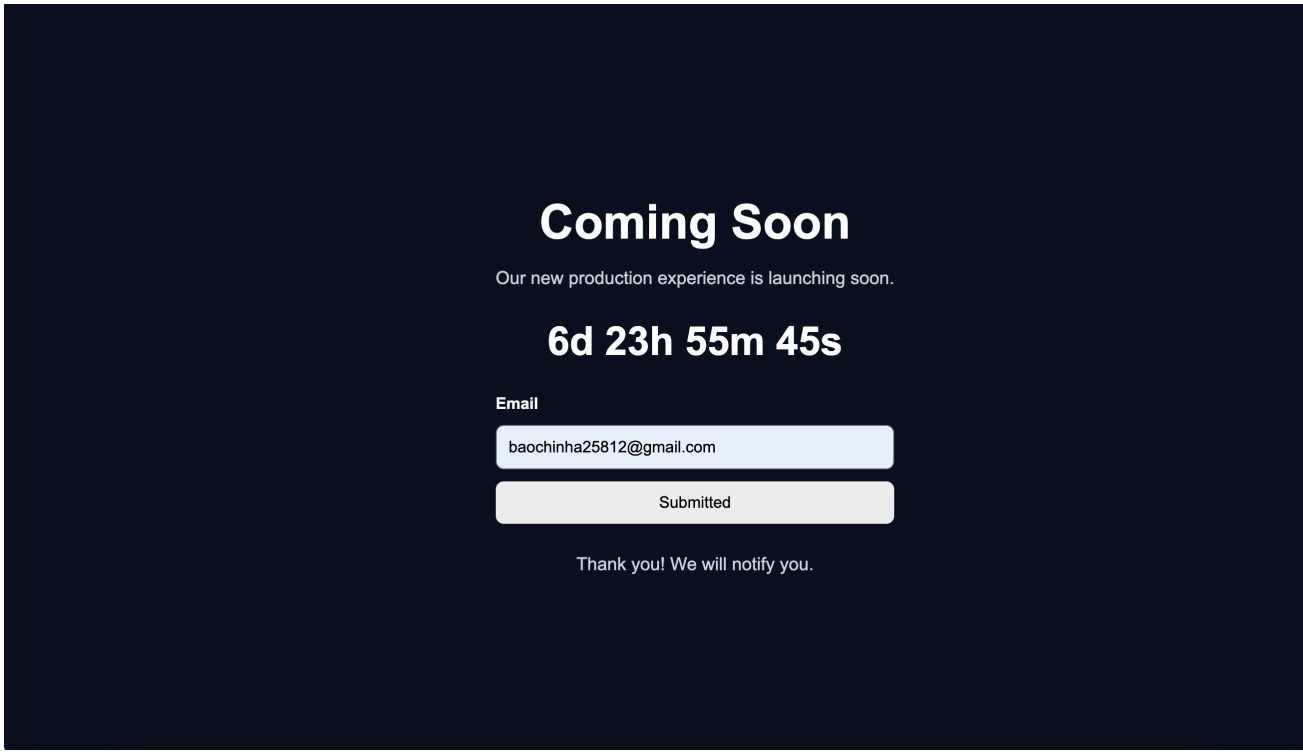
2. SLICE 2: State-Machine Form (Idle -> Submitting -> Success/Error).Sadasd


index.html
style.css
app.js
TASK_DECOMPOSITION.md
AI_FAILL
index.html > html > body > main > section.hero
Text Editor
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>HW3 Resilient Landing Page</title>
<link rel="stylesheet" href="style.css">
</head>
<body>
<main>
<section class="hero">
<h1>Coming Soon</h1>

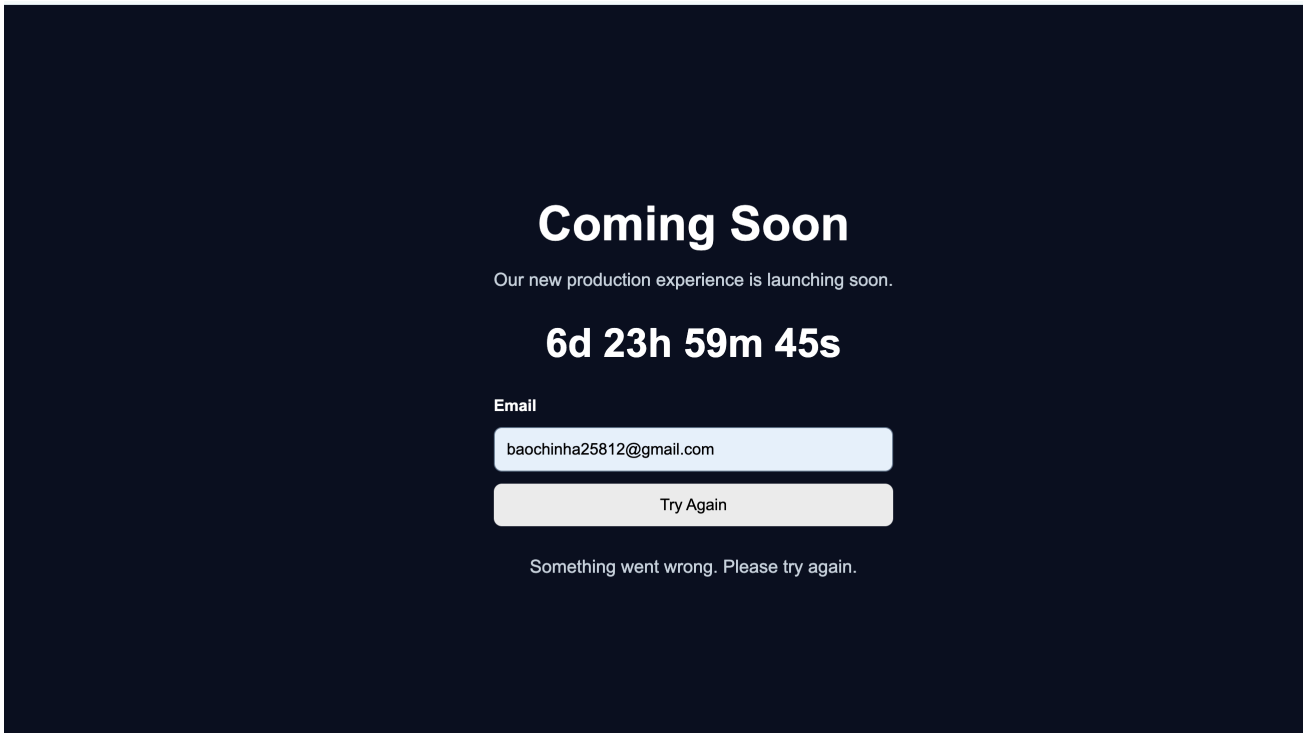
Our new production experience is launching soon.

Problems
Output
Debug Console
Terminal
Ports
SQL History
zsh
chinh@192 24520225_NguyenBaoChinh_Lab01_HW3 % git status
modified: index.html
modified: style.css
Untracked files:
(use "git add <file>..." to include in what will be committed)
AI_FAILURE_AUDIT.md
no changes added to commit (use "git add" and/or "git commit -a")
chinh@192 24520225_NguyenBaoChinh_Lab01_HW3 % git add index.html style.css app.js
chinh@192 24520225_NguyenBaoChinh_Lab01_HW3 % git commit -m "feat: implement form state machine"
[master 8e94049] feat: implement form state machine
3 files changed, 158 insertions(+), 1 deletion(-)
chinh@192 24520225_NguyenBaoChinh_Lab01_HW3 %

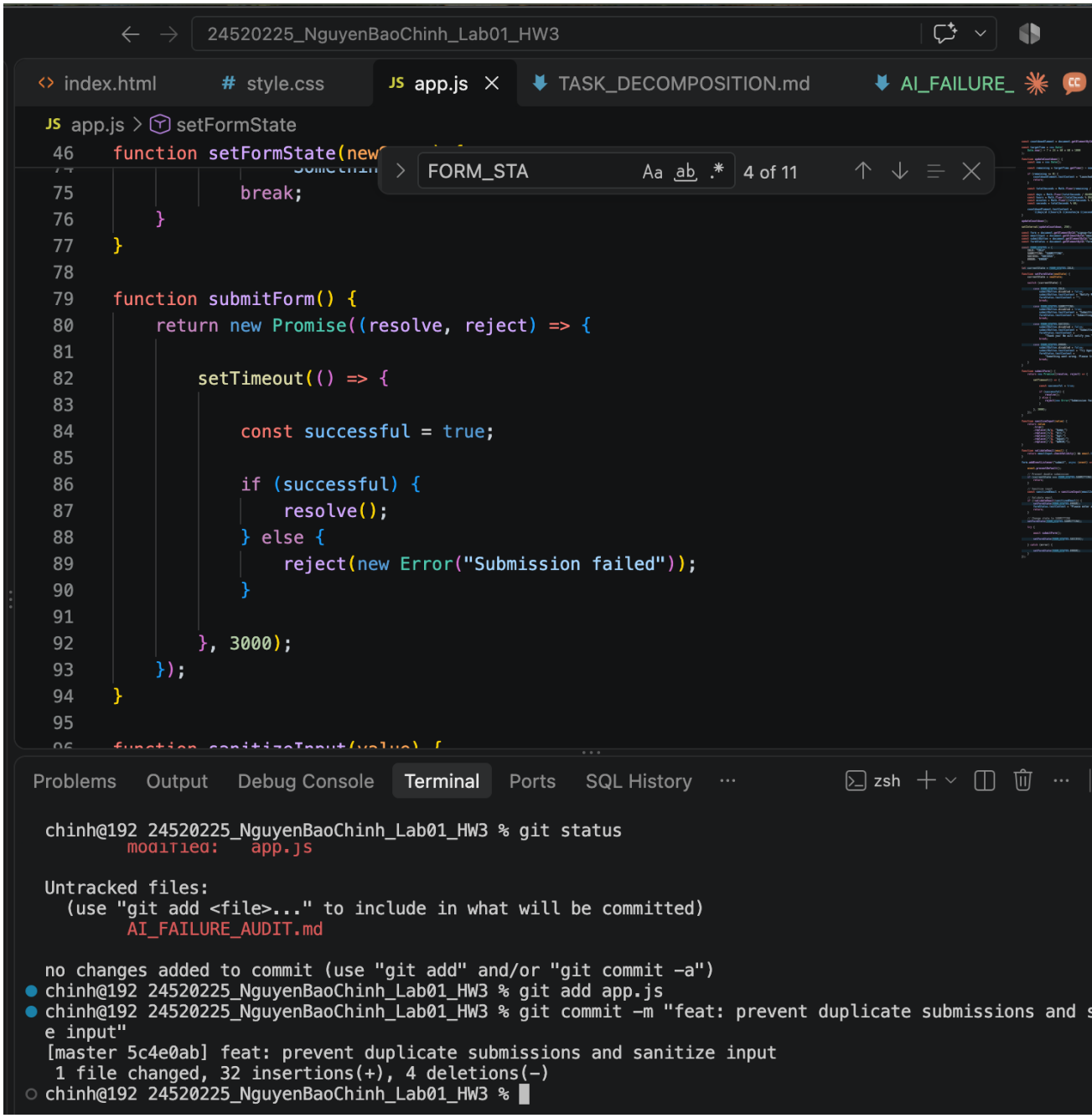
a. Sucess

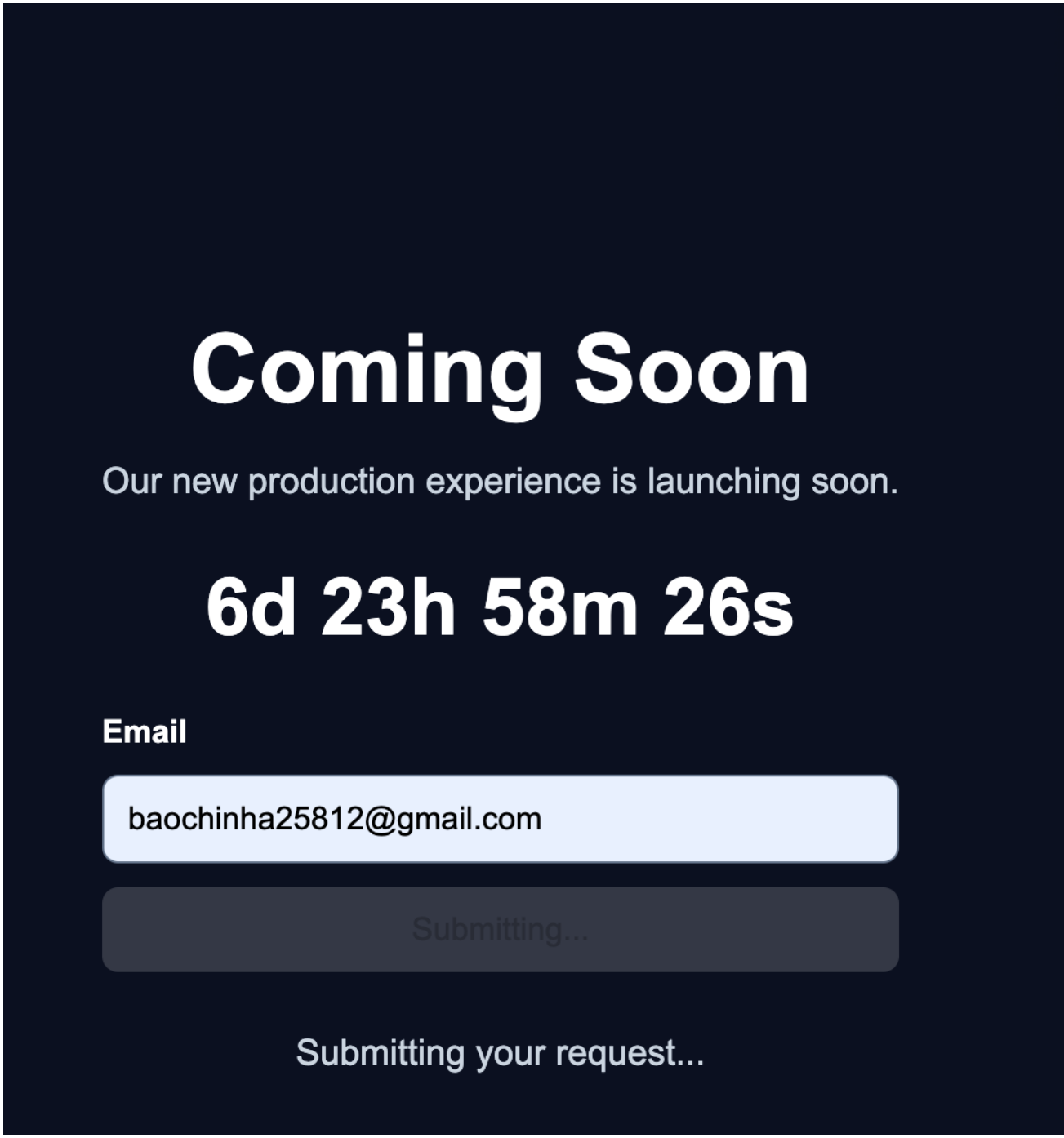


b. False



3. SLICE 3: Double-submit prevention & input sanitization (zero XSS).Adsasd


24520225_NguyenBaoChinh_Lab01_HW3
index.html
style.css
app.js
TASK_DECOMPOSITION.md
AI_FAILURE_
app.js > setFormState
FORM_STA
Aa ab .* 4 of 11
46 function setFormState(new
75 break;
76 }
77 }
78
79 function submitForm() {
80 return new Promise((resolve, reject) => {
81
82 setTimeout(() => {
83
84 const successful = true;
85
86 if (successful) {
87 resolve();
88 } else {
89 reject(new Error("Submission failed"));
90 }
91
92 }, 3000);
93 });
94 }
95
Problems
Output
Debug Console
Terminal
Ports
SQL History
zsh
chinh@192 24520225_NguyenBaoChinh_Lab01_HW3 % git status
modified: app.js
Untracked files:
(use "git add <file>..." to include in what will be committed)
AI_FAILURE_AUDIT.md
no changes added to commit (use "git add" and/or "git commit -a")
chinh@192 24520225_NguyenBaoChinh_Lab01_HW3 % git add app.js
chinh@192 24520225_NguyenBaoChinh_Lab01_HW3 % git commit -m "feat: prevent duplicate submissions and s
e input"
[master 5c4e0ab] feat: prevent duplicate submissions and sanitize input
1 file changed, 32 insertions(+), 4 deletions(-)
chinh@192 24520225_NguyenBaoChinh_Lab01_HW3 %


Coming Soon
Our new production experience is launching soon.
6d 23h 58m 26s
Email
baochinha25812@gmail.com
Submitting...
Submitting your request...